PassFrame tuition

Speed, Distance & Time

Year 4 Pt 2 — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

Lesson Worksheet — Answers

Easy / Standard

1. 70 km/h (140÷2=70)

2. B — 18×2=36

3. 50 km/h (250÷5=50)

4. 120 miles (60×2=120)

5. C — 110 km/h is a realistic motorway speed

6. 480 km/h (720÷1.5=480)

7. 15 km (6×2.5=15)

8. 55 km/h (165÷3=55)

9. A, B and D (120÷2=60, 90÷1.5=60 and 200÷2.5=80 are correct; 150÷3 is actually 50 (not 40) and 63÷3 is actually 21 (not 20))

10. 60 km/h (210÷3.5=60)

11. 7 km/h (21÷3=7)

12. C — The fastest speed (60 km/h) takes the least time

13. 130 km (65×2=130)

14. 70 km/h (175÷2.5=70)

15. 18 km/h (36 km ÷ 2 hours = 18 km/h)

Standard

1. 70 km/h (350÷5=70)

2. 42 km (12×3.5=42)

3. C — 260÷4=65

4. 40 km (30 min = 0.5 hours; 80×0.5=40)

5. 120 km/h (480÷4=120)

6. 110 km (44×2.5=110)

7. B — 550×2=1,100

8. 17 km/h (51÷3=17)

9. 25 km (15 min = 0.25 hours; 100×0.25=25)

10. 7.5 km (90 min = 1.5 hours; 5×1.5=7.5)

11. A, B, D and E (40×2=80; 40×0.5=20; 40×0.25=10; 40×1.5=60 are all correct; 40×3 is actually 120 (not 100), so that one is false)

12. 90 km/h (315÷3.5=90)

13. 130 km (52×2.5=130)

14. B — 140÷2=70 km/h, the highest of the four (60, 70, 63, 66)

15. 60 km/h (240 km ÷ 4 hours = 60 km/h)

Standard / Hard

1. 55 km/h (220 km ÷ 4 hours = 55 km/h)

2. B — 375÷75=5

3. 14 km/h (42 km ÷ 3 hours = 14 km/h)

4. 3 hours (1,050÷350=3)

6. B — Return takes 4 hours: 240÷4=60

7. 180 km (30×3=90, plus 45×2=90, total 180)

8. 360 km ((55+65)×3=360)

9. A, B, C and E (360÷60=6 hours; in 2 hours, 120 km; doubling speed halves the time to 3 hours; in 4 hours, 240 km — all correct; but half the journey (180 km) actually takes 180÷60=3 hours, not 4)

11. 6 hours (45÷15=3 hours, +1 hour rest, +40÷20=2 hours = 6 hours total)

12. B — 2.75 hours × 84 km/h = 231 km

13. 80 km/h (480 km ÷ 6 hours = 80 km/h)

14. 45 km ((28−10)×2.5=45)

5. 200 km (55×2=110, plus 60×1.5=90, total 200)

10. 6.5 hours (210÷70=3 hours, plus 210÷60=3.5 hours, total 6.5 hours)

15. 50 km/h (first part takes 280÷70=4 hours; remaining time=8−4=4 hours; 200÷4=50 km/h)

Homework — Answers

Easy / Standard

1. 60 km/h (90÷1.5=60)

2. B — Time = Distance ÷ Speed

3. 40 km (16×2.5=40)

4. 65 km/h (260÷4=65)

5. A — 5 km/h is a realistic walking speed

6. 9 km (6×1.5=9)

7. 70 km/h (105÷1.5=70)

8. 90 km (36×2.5=90)

9. A, C and D (150÷2.5=60, 126÷3=42 and 45÷1.5=30 are all correct; 80÷2 is actually 40 (not 50) and 200÷4 is actually 50 (not 40))

10. 55 km/h (220÷4=55)

11. 6 km/h (9÷1.5=6)

12. B — 54 km/h is the largest value

13. 115 km (46×2.5=115)

14. 70 km/h (245÷3.5=70)

15. 18 km/h (36 km ÷ 2 hours = 18 km/h)

Standard

1. 70 km/h (315÷4.5=70)

2. 35 km (14×2.5=35)

3. C — 310÷5=62

4. 54 km (45 min = 0.75 hours; 72×0.75=54)

5. 93 km/h (372÷4=93)

6. 95 km (38×2.5=95)

7. C — 620×1.5=930

8. 19 km/h (57÷3=19)

9. 30 km (20 min = 1/3 hour; 90×1/3=30)

10. 7 km (105 min = 1.75 hours; 4×1.75=7)

11. A, C, D and E (50×2=100; 50×3=150; 50×0.25=12.5; 1 hour 12 minutes = 1.2 hours, 50×1.2=60 — all correct; 30 minutes should give 25 miles, not 20, so that one is false)

12. 94 km/h (235÷2.5=94)

13. 145 km (58×2.5=145)

14. D — 144÷1.5=96 km/h, the highest of the four (70, 84, 77.5, 96)

15. 70 km/h (280 km ÷ 4 hours = 70 km/h)

Standard / Advanced

1. 180 minutes (165÷55=3 hours=180 minutes)

2. C — 2 hours 30 min = 2.5 hours; 175÷2.5=70

3. 480 km (45 min = 0.75 hours; 640×0.75=480)

4. 15 minutes (8÷32=0.25 hours=15 minutes)

6. 4.5 hours (120÷60=2 hours, plus 175÷70=2.5 hours, total 4.5 hours)

7. 2.5 hours (combined speed 160 km/h; 400÷160=2.5)

8. A, B, D and E (400÷80=5h; 400÷100=4h; 80 km/h means 20 km every 15 minutes; 400÷50=8h — all correct; but in 2 hours the car covers only 160 km, not half of 400 km (200 km), so that one is false)

9. 5 hours (48÷24=2 hours there, 48÷16=3 hours back, total 5 hours)

11. 38 km/h (total distance 190 km ÷ total time 5 hours = 38 km/h)

12. 2 hours 15 minutes (18÷12=1.5 hours plus 6÷8=0.75 hours = 2.25 hours = 2 hours 15 minutes)

13. C — The gap grows by 95−75=20 km/h; 20 × 2 = 40 km

14. 42.5 km/h (first part takes 280÷70=4 hours; remaining time=8−4=4 hours; 170÷4=42.5 km/h)

5. C — Speed = 65 km/h; 260÷65=4

10. C — Driving time = 225÷90=2.5 hours (2h30); plus 20-minute stop = 2h50; 07:40 + 2h50 = 10:30

15. 77 km/h (leg times: 1 + 1.5 + 0.67 ≈ 3.17 hours; total distance 245 km; 245 ÷ 3.17 ≈ 77.4, rounded to 77 km/h)